CHAPTER: TRANSACTIONS

CHAPTER 14: TRANSACTIONS

- Transaction Concept
- Transaction State
- Concurrent Executions
- Serializability
- Recoverability
- Implementation of Isolation

Transaction Definition in SQL

Testing for Serializability.

TRANSACTION CONCEPT

- A **transaction** is a *unit* of program execution that accesses and possibly updates various data items.

- E.g. transaction to transfer $50 from account A to account B:
  1. **read**(*A*)
  2. *A* := *A* – 50
  3. **write**(*A*)
  4. **read**(*B*)
  5. *B* := *B* + 50
  6. **write**(*B*)
- Two main issues to deal with:
  - Failures of various kinds, such as hardware failures and system crashes
  - Concurrent execution of multiple transactions

EXAMPLE OF FUND TRANSFER

- Transaction to transfer $50 from account A to account B:
  1. **read**(*A*)
  2. *A* := *A* – 50
  3. **write**(*A*)

4. **read**(B)
5. $B := B + 50$
6. **write**(B)

- **Atomicity requirement**
  - if the transaction fails after step 3 and before step 6, money will be "lost" leading to an inconsistent database state
    - Failure could be due to software or hardware
  - the system should ensure that updates of a partially executed transaction are not reflected in the database

EXAMPLE OF FUND TRANSFER (CONT.)

- **Durability requirement** — once the user has been notified that the transaction has completed (i.e., the transfer of the $50 has taken place), the updates to the database by the transaction must

persist even if there are software or hardware failures.

Example of Fund Transfer (Cont.)

- Transaction to transfer $50 from account A to account B:
  1. **read**(A)
  2. $A := A - 50$
  3. **write**(A)
  4. **read**(B)
  5. $B := B + 50$
  6. **write**(B)
- **Consistency requirement** in above example:
  - the sum of A and B is unchanged by the execution of the transaction

Example of Fund Transfer (Cont.)

- In general, consistency requirements include

Explicitly specified integrity constraints such as primary keys and foreign keys

- Implicit integrity constraints
  - e.g. sum of balances of all accounts, minus sum of loan amounts must equal value of cash-in hand

A transaction must see a consistent database.
During transaction execution the database may be temporarily inconsistent.

- When the transaction completes successfully the database must be consistent
  - Erroneous transaction logic can lead to inconsistency

Example of Fund Transfer (Cont.)

- **Isolation requirement** — if between steps 3 and 6, another transaction T2 is allowed to access the partially updated database, it will see an inconsistent database (the sum $A + B$ will be less than it should

be).**T1 T2**

1. **read**(*A*)
2. *A* := *A* – 50
3. **write**(*A*)
   read(A), read(B), print(A+B)
4. **read**(*B*)
5. *B* := *B* + 50
6. **write**(*B*

- Isolation can be ensured trivially by running transactions **serially**
  - that is, one after the other.
- However, executing multiple transactions concurrently has significant benefits, as we will see later.

ACID PROPERTIES

A **transaction** is a unit of program execution that accesses and possibly updates various data items. To preserve the integrity of data the database system must

ensure:

- **Atomicity.** Either all operations of the transaction are properly reflected in the database or none are.

- **Durability.** After a transaction completes successfully, the changes it has made to the database persist, even if there are system failures.

- **Consistency.** Execution of a transaction in isolation preserves the consistency of the database.

ACID Properties (Cont.)

- **Isolation.** Although multiple transactions may execute concurrently, each transaction must be unaware of other concurrently executing transactions.

Intermediate transaction results must be hidden from other concurrently executed transactions.

- That is, for every pair of transactions T_i and T_j, it appears to T_i that either T_j, finished execution before T_i started, or T_j started execution after T_i finished.

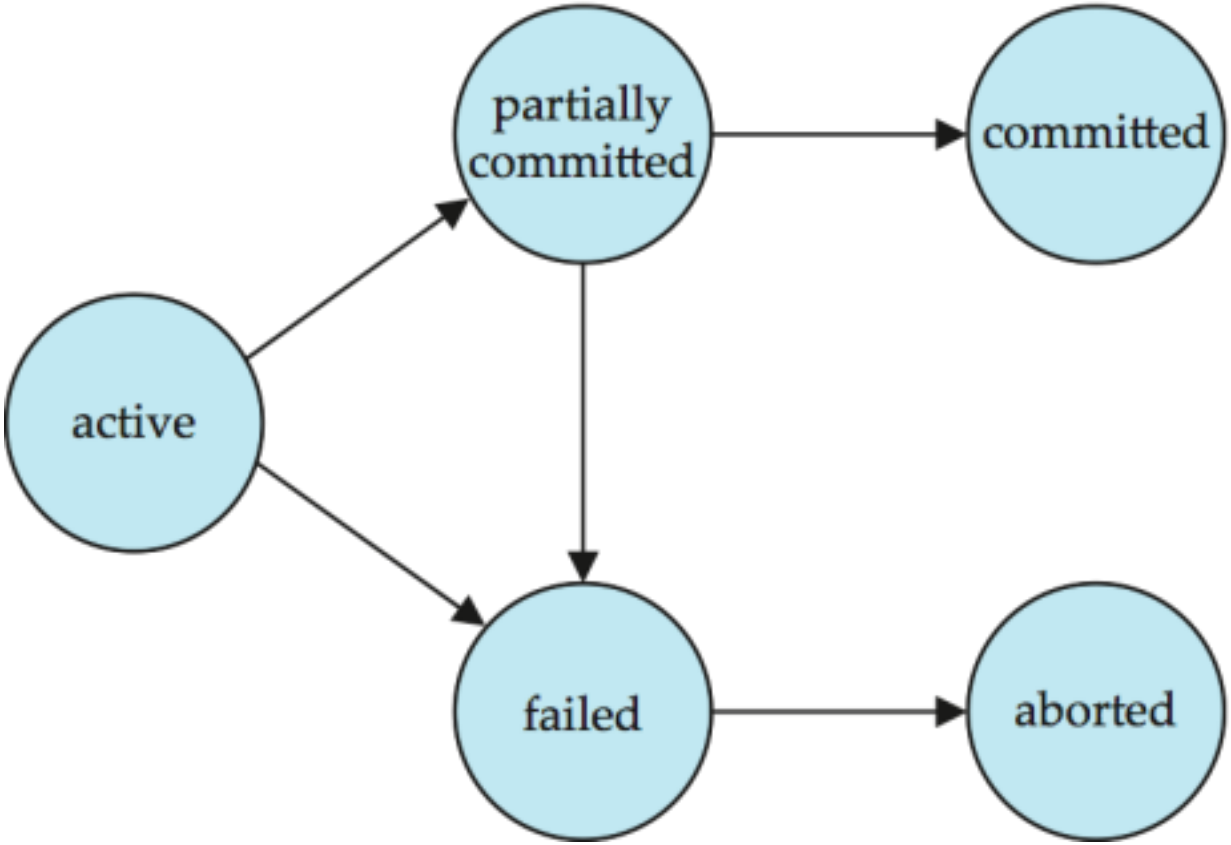
Transaction State

- **Active** – the initial state; the transaction stays in this state while it is executing
- **Partially committed** – after the final statement has been executed.
- **Failed** – after the discovery that normal execution can no longer proceed.
- **Aborted** – after the transaction has been rolled back and the database restored to its state prior to the start of the transaction. Two options after it has been aborted:

- restart the transaction
  - can be done only if no internal logical error
- kill the transaction

- **Committed** – after successful completion.

TRANSACTION STATE (CONT.)



Concurrent Executions

- Multiple transactions are allowed to run concurrently in the system. Advantages are:
  - **increased processor and disk utilization**, leading to better transaction *throughput*
    - E.g. one transaction can be using the CPU while another is reading from or writing to the disk
  - **reduced average response time** for transactions: short transactions need not wait behind long ones.

Concurrent Executions

- **Concurrency control schemes** – mechanisms to achieve isolation
  - that is, to control the interaction among the

concurrent transactions in order to prevent them from destroying the consistency of the database

- Will study in Chapter 16, after studying notion of correctness of concurrent executions.

SCHEDULES

- **Schedule** – a sequences of instructions that specify the chronological order in which instructions of concurrent transactions are executed
  - a schedule for a set of transactions must consist of all instructions of those transactions
  - must preserve the order in which the instructions appear in each individual transaction.

SCHEDULES

- A transaction that successfully completes its execution will have a commit instructions as the last statement
  - by default transaction assumed to execute commit instruction as its last step
- A transaction that fails to successfully complete its execution will have an abort instruction as the last statement

SCHEDULE 1

- Let T_1transfer \$50 from A to B, and T_2transfer 10% of the balance from A to B.
- A serial schedule in which $T_1$is followed by T_2:

T_1	T_2
read (A)	
$A := A - 50$	
write (A)	
read (B)	
$B := B + 50$	
write (B)	
commit	
	read (A)
	$temp := A * 0.1$
	$A := A - temp$
	write (A)
	read (B)
	$B := B + temp$
	write (B)
	commit

SCHEDULE 2

- A serial schedule where T_2 is followed by T_1

T_1	T_2
	read (A) $temp := A * 0.1$ $A := A - temp$ write (A) read (B) $B := B + temp$ write (B) commit
read (A) $A := A - 50$ write (A) read (B) $B := B + 50$ write (B) commit	

SCHEDULE 3

- Let T_1 and T_2 be the transactions defined previously. The following schedule is not a serial schedule, but it

is *equivalent* to Schedule 1.

T_1	T_2
read (A)	
$A := A - 50$	
write (A)	
	read (A)
	$temp := A * 0.1$
	$A := A - temp$
	write (A)
read (B)	
$B := B + 50$	
write (B)	
commit	
	read (B)
	$B := B + temp$
	write (B)
	commit

In Schedules 1, 2 and 3, the sum A + B is preserved.

SCHEDULE 4

- The following concurrent schedule does not preserve the value of $(A + B)$.

T_1	T_2
read (A)	
$A := A - 50$	
	read (A)
	$temp := A * 0.1$
	$A := A - temp$
	write (A)
	read (B)
write (A)	
read (B)	
$B := B + 50$	
write (B)	
commit	
	$B := B + temp$
	write (B)
	commit

SCHEDULE

Schedule

serial concurrent

Non

serializable Serialazable

View Conflict

SERIALIZABILITY

- A (possibly concurrent) schedule is serializable if it is equivalent to a serial schedule. Different forms of

schedule equivalence give rise to the notions of: 1. **conflict serializability**

2. **view serializability**

CONFLICTING INSTRUCTIONS(OPERATION)

- Instructions l_i and l_j of transactions T_i and T_j respectively, **conflict** if and only if there exists some item Q accessed by both l_i and l_j, and at least one of these instructions wrote Q.

1. l_i = **read**(Q), l_j = **read**(Q). l_i and l_j don't conflict.

2. l_i = **read**(Q), l_j = **write**(Q). They conflict.

3. l_i = **write**(Q), l_j = **read**(Q). They conflict

4. l_i = **write**(Q), l_j = **write**(Q). They conflict

Lost Update Problem (W-W Conflict)

- T1 T2
- R(X)
- X=X-10
- R(X)
- X=X+20 - W(X)
- R(Y)
- W(X) - Commit - Y=Y+10
- W(Y)
- Commit

Dirty Read Problem (W-R Conflict)

- T1 T2
- R(X)

- X=X-10
- W(X)

R(X)

- X=X+10 - W(X) - Commit

- ROLLBACK

Unrepeated Read Problem (R-W Conflict)

- T1 T2 - R(X)
- R(X) - X=X+10 - W(X)
-
- R(X)
-

Conflict Serializability

- We say that a schedule S is **conflict serializable** if it is conflict equivalent to a serial schedule.
- If a schedule S can be transformed into a schedule S' by a series of swaps of non-conflicting instructions, we say that S and S' are **conflict equivalent**.

Conflict Serializability (Cont.)

- Schedule 3 can be transformed into Schedule 6, a serial schedule where T_2 follows T_1, by series of swaps of non conflicting instructions. Therefore Schedule 3 is conflict serializable.

T_1	T_2
read (A)	
write (A)	
	read (A)
	write (A)
read (B)	
write (B)	
	read (B)
	write (B)

T_1	T_2
read (A)	
write (A)	
read (B)	
write (B)	
	read (A)
	write (A)
	read (B)
	write (B)

Schedule 3 Schedule 6

CONFLICT SERIALIZABILITY (CONT.)

- Example of a schedule that is not conflict serializable:

T_3	T_4
read (Q)	
	write (Q)
write (Q)	

- We are unable to swap instructions in the above schedule to obtain either the serial schedule < T_3, T_4 >, or the serial schedule < T_4, T_3 >.

View Serializability

- Let S and S' be two schedules with the same set of transactions. S and S' are **view equivalent** if the following three conditions are met, for each data item Q,
  1. If in schedule S, transaction T_i reads the initial value of Q, then in schedule S' also transaction T_i must read the initial value of Q.
  2. If in schedule S transaction T_i executes **read**(Q), and that

value was produced by transaction T_j(if any), then in schedule S' also transaction T_i must read the value of Q that was produced by the same **write**(Q) operation of transaction T_j.

3. The transaction (if any) that performs the final **write**(Q) operation in schedule S must also perform the final **write**(Q) operation in schedule S'.

As can be seen, view equivalence is also based purely on **reads** and **writes** alone.

VIEW SERIALIZABILITY (CONT.)

- A schedule S is **view serializable** if it is view equivalent to a serial schedule.
- Every conflict serializable schedule is also view serializable.

Below is a schedule which is view-serializable but *not* conflict serializable.

T_{27}	T_{28}	T_{29}
read (Q)		
	write (Q)	
write (Q)		
		write (Q)

- What serial schedule is above equivalent to?
- Every view serializable schedule that is not conflict serializable has **blind writes**.

Other Notions of Serializability

- The schedule below produces same outcome as the serial schedule $< T_1, T_5 >$, yet is not conflict equivalent or view equivalent to it.

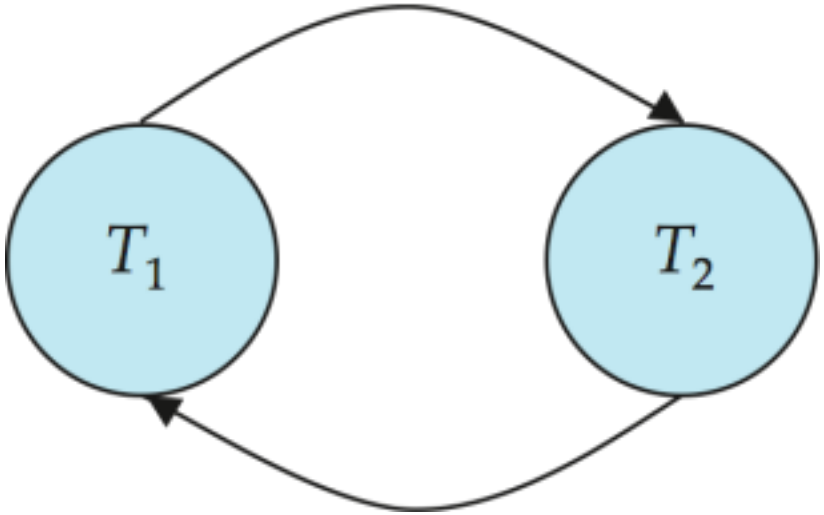
T_1	T_5
read (A)	
$A := A - 50$	
write (A)	
	read (B)
	$B := B - 10$
	write (B)
read (B)	
$B := B + 50$	
write (B)	
	read (A)
	$A := A + 10$
	write (A)

- Determining such equivalence requires analysis of operations other than read and write.

TESTING FOR SERIALIZABILITY

- Consider some schedule of a set of transactions $T_1, T_2, ..., T_n$
- **Precedence graph** — a directed graph where the vertices are the transactions (names). We draw an arc from T_i to T_j if the two transaction

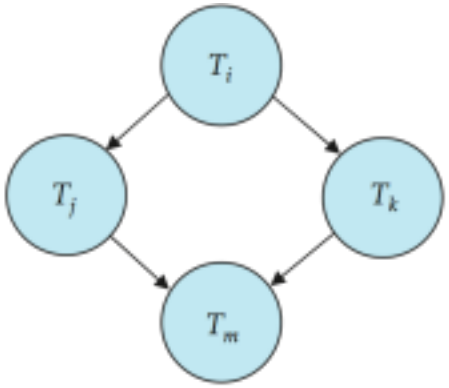
conflict, and T_i accessed the data item on which the conflict arose earlier. We may label the arc by the item that was accessed.



- **Example 1**

TEST FOR CONFLICT SERIALIZABILITY

- A schedule is conflict serializable if and only if its precedence graph is acyclic.
  - Cycle-detection algorithms exist which take order n^2 time, where n is the number of vertices in the graph.



(a)

- (Better algorithms take order $n + e$ where e is the number of edges.)

- If precedence graph is acyclic, the serializability order can be obtained by a *topological sorting* of the graph.
  - This is a linear order consistent with the partial order of the graph.
  - For example, a serializability order for Schedule A would be
    $T_5 \rightarrow T_1 \rightarrow T_3 \rightarrow T_2 \rightarrow T_4$
    - Are there others?

Recoverable Schedules

- **Recoverable schedule** — if a transaction T_j reads a data item previously written by a transaction T_i, then the commit operation of T_i appears before the commit operation of T_j.
- The following schedule (Schedule 11) is not recoverable if T_9 commits immediately after the read

T_8	T_9
read (A)	
write (A)	
	read (A)
	commit
read (B)	

- If T_8 should abort, T_9 would have read (and possibly shown to the user) an inconsistent database state. Hence, database must ensure that schedules are recoverable.

Cascading Rollbacks

- **Cascading rollback** – a single transaction failure leads to a series of transaction rollbacks. Consider the following schedule where none of the transactions has yet committed (so the schedule is recoverable)

T_{10}	T_{11}	T_{12}
read (A)		
read (B)		
write (A)		
	read (A)	
	write (A)	
		read (A)
abort		

If T_{10} fails, T_{11} and T_{12} must also be rolled back. Can lead to the undoing of a significant amount of work